CIE International A Level Physics

# Topic 18

# Electric Fields

Revision Booklet

**This booklet covers:**

- Electric Fields and Field Lines
- Uniform Electric Fields
- Coulomb's Law
- Electric Field of a Point Charge
- Electric Potential

## Core Concepts and Definitions

### Electric Field

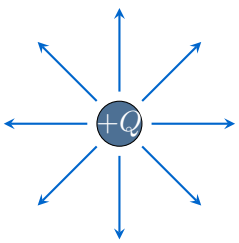
An **electric field** is a region of space in which a charged object experiences a force.

- Electric field is defined as the **force per unit positive charge** acting on a small stationary test charge placed at that point.
- It is a **vector** quantity, directed along the force on a positive test charge.
- Units: N $C^{-1} \equiv$ V $m^{-1}$.

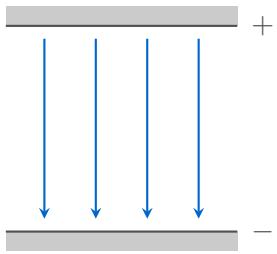
$$E = \frac{F}{q}$$

### Radial vs Uniform Fields

- **Radial field** (around a point charge or sphere): field lines point radially inward (negative charge) or outward (positive charge); $E$ decreases with distance.
- **Uniform field** (between parallel plates): field lines are parallel and equally spaced; $E$ is constant throughout.




**Radial Field (+)**




**Uniform Field**

## Uniform Electric Fields

### Field Strength Between Parallel Plates

$$E = \frac{\Delta V}{\Delta d}$$

$E$ = electric field strength (V $m^{-1}$)
$\Delta V$ = potential difference between the plates (V)
$\Delta d$ = separation of the plates (m)

### Motion of Charged Particles in a Uniform Field

- A charge $q$ in a uniform field $E$ experiences a constant force $F = qE$.
- The motion is analogous to projectile motion in a gravitational field:
  - Along the field: **uniform acceleration** $a = qE/m$.

  - Perpendicular to the field: **constant velocity** (if no other forces).
- The path of the charge is **parabolic**.

**Common Mistake**

$E = \Delta V/\Delta d$ applies **only** to uniform fields (parallel plates). Do not apply it to the field around a point charge — use $E = Q/(4\pi\varepsilon_0 r^2)$ instead.

## Coulomb's Law

**Coulomb's Law**

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$$

| | |
|---|---|
| $F$ | = electrostatic force between the charges (N) |
| $Q_1$, $Q_2$ | = the two point charges (C) |
| $r$ | = separation between the charges (m) |
| $\varepsilon_0$ | = permittivity of free space = $8.85 \times 10^{-12}$ F m$^{-1}$ |

**Key Points**

- Like charges **repel**; unlike charges **attract**.
- The force obeys an **inverse-square law**: double the distance, quarter the force.
- Applies strictly to **point charges**, and to uniform spheres (treat as point charge at centre).
- Compare with gravity: same inverse-square form but gravity is **always attractive**.

**Comparison: Coulomb's Law vs Newton's Law of Gravitation**

| **Coulomb's Law** | **Newton's Law** |
|---|---|
| $F = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$ | $F = \dfrac{G m_1 m_2}{r^2}$ |
| Can be attractive or repulsive | Always attractive |
| Acts between charges | Acts between masses |
| Much stronger force | Much weaker force |

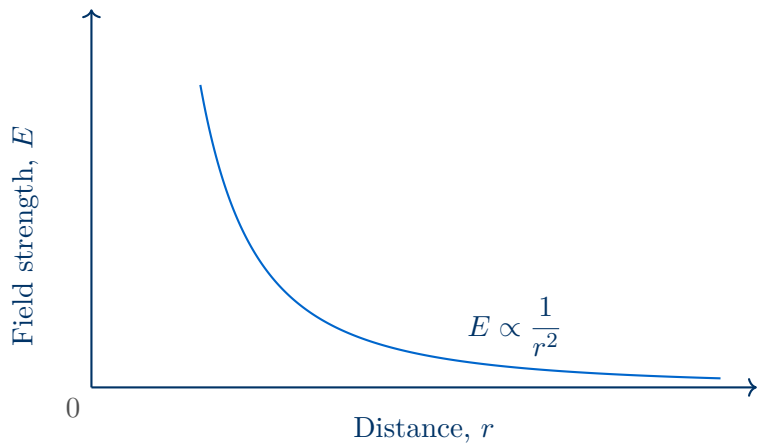
## Electric Field of a Point Charge

### Electric Field Strength due to a Point Charge

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

$E$ = electric field strength at distance $r$ (N C$^{-1}$)
$Q$ = point charge creating the field (C)
$r$ = distance from the centre of $Q$ (m)
$\varepsilon_0$ = $8.85 \times 10^{-12}$ F m$^{-1}$

**Variation of $E$ with distance $r$**




### Comparing $E$ and $g$ fields

- Both follow an inverse-square law with distance.
- $E = Q/(4\pi\varepsilon_0 r^2)$ parallels $g = GM/r^2$.
- $F = qE$ parallels $F = mg$.
- Unlike gravitational fields, electric fields can point inward or outward depending on the sign of $Q$.

## Electric Potential

### Definition of Electric Potential $V$

The **electric potential** at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point.

$$V = \frac{W}{q} \qquad \text{units: J C}^{-1} \equiv \text{V}$$

- $V = 0$ at infinity (by convention).
- Around a positive charge: $V > 0$ (work must be done *against* repulsion).

- Around a negative charge: $V < 0$ (work is done *by* the field as the test charge moves in).

**Electric Potential due to a Point Charge**

$$V = \frac{Q}{4\pi\varepsilon_0 r}$$

$V$ = electric potential at distance $r$ (V)
$Q$ = point charge (C)
$r$ = distance from the charge (m)

## Relationship Between $E$ and $V$

**Field Strength from Potential Gradient**

$$E = -\frac{\Delta V}{\Delta r}$$

The electric field strength equals the **negative gradient** of the potential–distance graph. The area under an $E$–$r$ graph gives the change in potential $\Delta V$.

## Electric Potential Energy

**Potential Energy of Two Point Charges**

$$E_P = \frac{Qq}{4\pi\varepsilon_0 r}$$

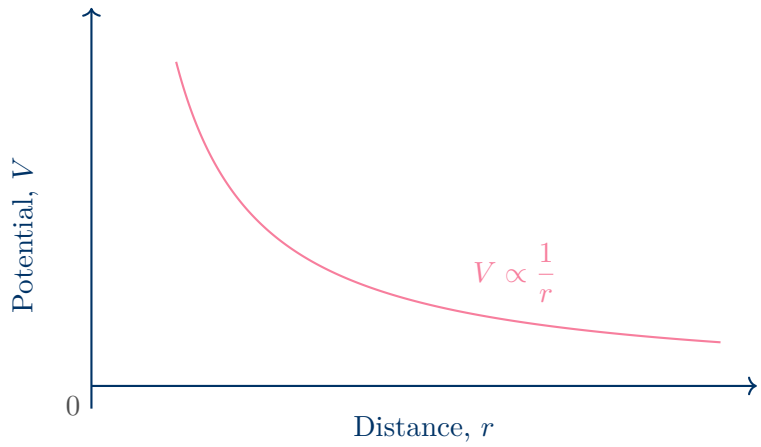
$E_P$ = electric potential energy (J)
$Q$, $q$ = the two point charges (C)
$r$ = separation (m)

Note: $E_P = qV$, where $V$ is the potential due to charge $Q$ at the location of $q$.

**Signs of Potential Energy**

- Like charges ($Qq > 0$): $E_P > 0$ — energy must be supplied to bring them together.
- Unlike charges ($Qq < 0$): $E_P < 0$ — energy is released as they come together.
- $E_P = 0$ at infinite separation.

**Graphs of $V$ and $E$ against $r$ for a positive point charge**




**Common Mistake**

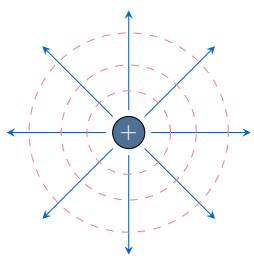
Note the difference in distance dependence: $E \propto 1/r^2$ but $V \propto 1/r$. Students often mix these up. Remember: potential falls off more *slowly* than field strength with distance.

## Equipotential Surfaces

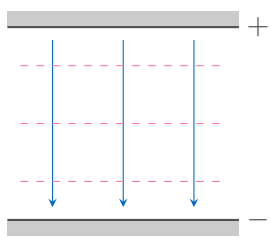
**Equipotentials**

An **equipotential surface** is a surface on which the electric potential is the same at every point.

- No work is done moving a charge *along* an equipotential.
- Equipotentials are always **perpendicular** to field lines.
- Around a point charge: equipotentials are concentric spheres.
- Between parallel plates: equipotentials are parallel planes equally spaced (for uniform field).



**Point charge**




**Parallel plates**

## Formula Summary Sheet

| Formula | Quantity | Units |
|---|---|---|
| $E = \dfrac{F}{q}$ | Electric field strength (definition) | N C$^{-1}$ |
| $F = qE$ | Force on a charge in a field | N |
| $E = \dfrac{\Delta V}{\Delta d}$ | Field between parallel plates (uniform) | V m$^{-1}$ |
| $F = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$ | Coulomb's Law | N |
| $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$ | Field due to a point charge | N C$^{-1}$ |
| $V = \dfrac{Q}{4\pi\varepsilon_0 r}$ | Potential due to a point charge | V |
| $E = -\dfrac{\Delta V}{\Delta r}$ | Field from potential gradient | V m$^{-1}$ |
| $E_P = \dfrac{Qq}{4\pi\varepsilon_0 r}$ | Electric potential energy | J |
| $E_P = qV$ | Potential energy of charge in field | J |

**Constants:** $\varepsilon_0 = 8.85\times10^{-12}$ F m$^{-1}$, $\dfrac{1}{4\pi\varepsilon_0} = 8.99\times10^{9}$ N m$^2$C$^{-2}$, $e = 1.60\times10^{-19}$ C

**Note:** $E \propto 1/r^2$ (inverse-square) but $V \propto 1/r$ (inverse).

## Worked Examples

### Example 1 — Force Between Two Charges

**Question:** Calculate the electrostatic force between two charges of $+3.0\ \mu$C and $-5.0\ \mu$C separated by 0.12 m.

**Solution**

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2} = \frac{3.0\times10^{-6}\times5.0\times10^{-6}}{4\pi\times8.85\times10^{-12}\times(0.12)^2}$$

$$F = \frac{1.5\times10^{-11}}{1.61\times10^{-12}} = \mathbf{9.3}\ \text{N}$$

The force is **attractive** (unlike charges).

### Example 2 — Field Strength and Potential at a Distance

**Question:** A point charge $Q = +4.0\ \mu$C. Calculate (a) the electric field strength and (b) the electric potential at a distance of 0.30 m.

**Solution**

**(a)** $$E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{4.0 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times (0.30)^2}$$

$$E = \frac{4.0 \times 10^{-6}}{1.005 \times 10^{-10}} \times (0.09)^{-1}$$

$$E = \frac{4.0 \times 10^{-6}}{1.005 \times 10^{-10}} = \mathbf{4.0 \times 10^5}\ \text{N C}^{-1}$$

**(b)** $$V = \frac{Q}{4\pi\varepsilon_0 r} = \frac{4.0 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times 0.30} = \mathbf{1.2 \times 10^5}\ \text{V}$$

### Example 3 — Uniform Field Between Plates

**Question:** Two parallel plates are separated by 4.0 mm with a potential difference of 240 V. Calculate (a) the field strength and (b) the force on an electron between the plates.

**Solution**

**(a)** $$E = \frac{\Delta V}{\Delta d} = \frac{240}{4.0 \times 10^{-3}} = \mathbf{6.0 \times 10^4}\ \text{V m}^{-1}$$

**(b)** $$F = qE = 1.6 \times 10^{-19} \times 6.0 \times 10^4 = \mathbf{9.6 \times 10^{-15}}\ \text{N}$$

## Practice Exam Questions

### Section A — Short Answer Questions

**Q1.** Define electric field strength and state its SI units.

*[2 marks]*

**Q2.** State Coulomb's Law and identify one similarity and one difference compared with Newton's Law of Gravitation.

*[3 marks]*

**Q3.** Define electric potential at a point. Explain why the potential around an isolated positive charge is positive, but around an isolated negative charge is negative.

*[3 marks]*

**Q4.** Two parallel plates are 6.0 mm apart. The electric field strength between them is $5.0 \times 10^4$ V m$^{-1}$. Calculate the potential difference between the plates.

*[2 marks]*

**Q5.** A point charge produces an electric field of $3.6 \times 10^5$ N C$^{-1}$ at a distance of 0.10 m. Calculate the magnitude of the charge.

*[3 marks]*

## Section B — Longer Structured Questions

**Q6.** Two point charges $Q_1 = +6.0\ \mu$C and $Q_2 = +6.0\ \mu$C are placed 0.20 m apart.

(a) Calculate the force between the two charges.

*[2 marks]*

(b) Calculate the electric field strength at the midpoint between the two charges. Explain your reasoning.

*[3 marks]*

(c) Calculate the electric potential at the midpoint between the two charges.

*[3 marks]*

**Q7.** An electron (mass $9.11 \times 10^{-31}$ kg, charge $-1.6 \times 10^{-19}$ C) enters horizontally between two parallel plates. The plates are 30 mm apart and have a potential difference of 150 V. The electron enters midway between the plates with horizontal speed $4.0 \times 10^{7}$ m s$^{-1}$.

(a) Calculate the electric field strength between the plates.

*[1 mark]*

(b) Calculate the vertical acceleration of the electron.

*[2 marks]*

(c) The plates are 60 mm long. Calculate the vertical deflection of the electron as it exits the plates.

*[3 marks]*

(d) State and explain whether the electron hits one of the plates before exiting.

*[2 marks]*

## Mark Scheme and Answers

**Q1.** Electric field strength is the force per unit positive charge acting on a small stationary test charge placed at that point [1]; units: N $C^{-1}$ or V $m^{-1}$ [1].

**Q2.** The force between two point charges is proportional to the product of the charges and inversely proportional to the square of their separation [1]. Similarity: both follow an inverse-square law [1]. Difference: gravity is always attractive; electrostatic force can be attractive or repulsive [1].

**Q3.** Electric potential at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point [1]. Around a positive charge: the test charge is repelled, so work must be done *on* it to bring it in from infinity — the potential is positive [1]. Around a negative charge: the test charge is attracted, so work is done *by* the field — the potential is negative [1].

**Q4.** $\Delta V = E \times \Delta d = 5.0 \times 10^4 \times 6.0 \times 10^{-3} = \mathbf{300}$ V [2].

**Q5.** $E = Q/(4\pi\varepsilon_0 r^2)$ [1]; $Q = E \times 4\pi\varepsilon_0 r^2 = 3.6 \times 10^5 \times 4\pi \times 8.85 \times 10^{-12} \times (0.10)^2$ [1] $= \mathbf{4.0 \times 10^{-7}}$ C [1].

**Q6(a).** $F = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 r^2} = \dfrac{(6.0 \times 10^{-6})^2}{4\pi \times 8.85 \times 10^{-12} \times (0.20)^2}$ [1] $= \mathbf{8.1}$ N [1].

**Q6(b).** By symmetry, the two equal charges produce equal and opposite field contributions at the midpoint [1]; the fields cancel, so $E = \mathbf{0}$ at the midpoint [2].

**Q6(c).** $V_1 = V_2 = \dfrac{Q}{4\pi\varepsilon_0 r} = \dfrac{6.0 \times 10^{-6}}{4\pi \times 8.85 \times 10^{-12} \times 0.10}$ [1] $= 5.4 \times 10^5$ V; total $V = V_1 + V_2 = \mathbf{1.08 \times 10^6}$ V [2].

**Q7(a).** $E = \Delta V/\Delta d = 150/(30 \times 10^{-3}) = \mathbf{5.0 \times 10^3}$ V $m^{-1}$ [1].

**Q7(b).** $F = qE = 1.6 \times 10^{-19} \times 5.0 \times 10^3 = 8.0 \times 10^{-16}$ N [1]; $a = F/m = 8.0 \times 10^{-16}/(9.11 \times 10^{-31}) = \mathbf{8.8 \times 10^{14}}$ m $s^{-2}$ [1].

**Q7(c).** Time in field: $t = L/v = 0.060/(4.0 \times 10^7) = 1.5 \times 10^{-9}$ s [1]; vertical deflection: $y = \frac{1}{2}at^2 = \frac{1}{2} \times 8.8 \times 10^{14} \times (1.5 \times 10^{-9})^2$ [1] $= \mathbf{9.9 \times 10^{-4}}$ m $\approx \mathbf{1.0}$ mm [1].

**Q7(d).** The electron enters midway so has 15 mm to the nearest plate [1]; deflection of $\approx 1$ mm $\ll 15$ mm, so the electron **does not** hit a plate [1].

## Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

| | Learning Objective | Confidence (1–3) |
|---|---|---|
| ☐ | Define electric field strength as force per unit positive charge; use $E = F/q$ | |
| ☐ | Represent electric fields using field lines for point charges and parallel plates | |
| ☐ | Use $E = \Delta V/\Delta d$ for a uniform field between parallel plates | |
| ☐ | Describe the parabolic motion of a charge in a uniform electric field | |
| ☐ | State and apply Coulomb's Law $F = Q_1Q_2/(4\pi\varepsilon_0 r^2)$ | |
| ☐ | Use $E = Q/(4\pi\varepsilon_0 r^2)$ for the field due to a point charge | |
| ☐ | Define electric potential; explain why sign depends on sign of source charge | |
| ☐ | Use $V = Q/(4\pi\varepsilon_0 r)$ for potential due to a point charge | |
| ☐ | Apply $E = -\Delta V/\Delta r$ to relate field strength and potential gradient | |
| ☐ | Use $E_P = Qq/(4\pi\varepsilon_0 r)$ for potential energy of two charges | |
| ☐ | Sketch and interpret equipotential diagrams for point charges and parallel plates | |
| ☐ | Compare electric and gravitational fields (similarities and differences) | |

*Key: 1 = Need more work 2 = Getting there 3 = Confident*

**Good luck with your revision!**

The key to electric fields is seeing the deep parallel with gravitational fields: same inverse-square laws, same potential formalism — but with charge replacing mass, and the crucial difference that charge can be positive or negative.